YILDIZ TECHNICAL UNIVERSITY
FACULTY OF ELECTRICAL AND ELECTRONICS ENGINEERING

Multidisciplinary Design Project (MDP)



TWO-WHEELED INDOOR PATIENT ASSISTANCE ROBOT

Wearable command interface + automatic charging station + mobile visual evaluation

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Project essence:
The two-wheeled robot actively maintains its own balance, carries a medication box (e.g., aspirin) to the target point using wearable-interface commands, proceeds to the automatic charging station at the end of the task, and begins charging without human intervention. The smartphone application monitors the task using the phone's own camera and produces a results report.

Document type: Project definition + technical specification + recommended division of work + 5×20-point evaluation rubric

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1. Project Scenario and Real-World Relevance

The project is based on an assistance-robot scenario for delivering small, pre-prepared, safe medication packages indoors to individuals receiving home care or having limited mobility. The robot can be directed by a caregiver or user through a wearable interface; after carrying its load to the target point, it proceeds to the charging station and starts charging without human intervention.

In the demonstration scenario, the robot carries a medication box (e.g., aspirin) as the standard task load. The robot does not select medication, determine dosage, or administer medication; it only represents the safe transport of a pre-prepared box from one indoor location to another. The smartphone application monitors and measures the robot's task performance using the phone's own camera and sensors and generates a results report.

Design principle
Results, not methods, are evaluated. The control algorithm, microcontroller, sensor, mechanical design, programming language, and manufacturing method are left to the team; however, all teams are subject to the same minimum requirements and the same physical acceptance tests.

2. Mandatory Project Modules

Code	Module	Minimum output
A	Balancing robot	Active balancing and controlled motion on two coaxial main wheels without external support.
B	Wearable command interface	Wireless generation of forward/backward/right/left/stop commands from an IMU, EMG, or equivalent human input on the glove.
C	Automatic charging station	Autonomous docking of the robot with the charging station without human or wearable-interface guidance; automatic start of charging without a person connecting a cable, operating a switch, or manually placing the robot on the station.
D	Mobile monitoring and evaluation	Reporting the robot's charging status through the phone application; displaying the current tilt angle using the robot's sensors; detecting robot motion through the phone camera to track and report task execution.
E	Medication-box transport	Safe transport of the standard medication box on the robot throughout the task and delivery to the target point.

3. Recommended Project Management and Balanced Division of Work

The following task allocation is a recommendation. Depending on team composition and advisor preference, it may be adopted as is, rearranged, or the division of work may be determined by the team. The recommended structure assigns each of the five departments a principal responsibility of comparable scope, aligned with its discipline and directly affecting system success.

Department	Recommended primary responsibility	Concrete deliverables
Control and Automation Engineering	Balance, motion, and station-approach control	Robot dynamics and state estimation; balance controller; forward/backward and turning motions; stability under load; controlled approach to and stopping at the station.
Electrical Engineering	Energy, power distribution, and automatic charging station	Battery and power architecture; protection elements; charging-station power stage; robot-station energy connection; automatic charging initiation and hardware charging-status indicator.
Electronics and Communications Engineering	Wearable electronics and wireless communication	Microcontroller/sensor electronics on the glove; wireless data link; robot-side receiver; communication reliability; interface boards and signal integrity.

Department	Recommended primary responsibility	Concrete deliverables
Biomedical Engineering	Human-machine interface, wearable sensor, and ergonomics	Selection and placement of IMU/EMG or equivalent human input; user calibration; biological/biomechanical signal processing; conversion into five commands; ergonomics and safety of the wearable system.
Computer Engineering	Mobile evaluation application and system software	Establishing communication between the smartphone application and the robot to receive and visualize charging and tilt information; detecting robot motion with the phone camera for task tracking; reporting task status.

Note: The division of work is not mandatory; it is a balanced starting recommendation. If the team structure differs, responsibilities may be reassigned by the advisor; however, all mandatory modules must be completed and each student's concrete contribution must be stated in the report.

4. Technical Specification

4.1 Fixed Minimum Conditions for All Teams

Constraint	Definitive requirement
Mechanical arrangement	In scored tests, the robot balances only on its two main wheels. A third wheel, foot, or any support continuously contacting the floor may not be used.
Active balancing	The primary actuator for balancing is the drive of the two main wheels. When unpowered, the robot may not be designed to remain passively upright or return itself to the upright position.
Passive-stability pretest	With the motors unpowered, when the robot is separately released from +10° and -10° body tilt, it must not return to the upright position within 5 s or remain passively upright. If this condition is not met, the system is not accepted as an "actively balanced robot."
Power	The robot operates from its own energy source during scored tests; the main supply is at most 24 V DC. During charging, human connection of a cable or manual intervention in the power connection is not permitted.
Safety	An accessible physical motor stop/main power cutoff, a protected battery pack/BMS for lithium batteries, and a fuse or electronic current protection in the main motor line are mandatory.
Ready-made solution	A complete working self-balancing robot/kit may not be used as the main platform. General-purpose boards, sensors, drivers, motors, charging modules, and mechanical parts may be used.
Load	The standard task load is one medication box (e.g., aspirin). The box must be transported closed, dry, unbreakable, and spill-proof; it must remain securely on the robot throughout the scored task but must not be permanently mounted to the body.
Evaluation integrity	After the final evaluation has started, firmware, control parameters, mechanical structure, or wiring may not be changed; only restarting and charging/replacing with the same type of battery are allowed.

4.2 Areas Left to Team Discretion

- Control method: PID, LQR, MPC, fuzzy/adaptive/sliding-mode control, optimization, learning-based methods, or a hybrid approach.
- Dimensions, body geometry, material, exterior design, manufacturing method, and load-carrier design.
- Brand/model of the microcontroller/SBC, IMU/EMG, motor, motor driver, communication infrastructure, charging technology, and sensors.
- The gesture/signal type used in the wearable interface and the mapping of the five commands to user movements.
- The mobile application development environment, image-processing algorithm, and report interface.
- Additional features such as automatic departure from the station after charging is complete, extra sensors, telemetry, sound/light, mapping, AI, etc., provided that the minimum requirements are not compromised.

4.3 Automatic Charging Station and Verification

The robot must reach the defined approach region of the charging station under its own motion or through wearable-glove guidance, and then autonomously dock with the charging station without user intervention. After correct docking, charging must start automatically without any person connecting a cable, operating a switch, or manually positioning the robot.

Item	Mandatory verification
Autonomous docking	The charging station remains fixed in the evaluation area. When the robot is placed at the starting position within the defined 90° approach region of a 30 cm-radius area centered on the charging station, from that point onward it must detect the charging station using an IR transmitter-receiver sensor without any human or wearable-interface guidance. The robot must adjust its own heading and forward speed, approach the station, perform the alignment required for charging, and dock in the charging position. Manually lifting or pushing the robot into the station is not permitted.
Automatic charging start	When the robot reaches the correct position, charging must start automatically within 5 s. In charging mode, after autonomous docking is complete, it may rest against the dock or be secured mechanically/magnetically. Once stable, the motors must be stopped. Energy transfer may use contact pads, spring contacts, or an inductive method; the user may not connect a cable.
Charging indicator	A visible charging indicator must be present on the robot or station. The indicator must be driven by the charging circuit's CHG/STAT output or an equivalent hardware charging-status signal and must not be independently activated by software.
Simple verification	The indicator is on while the robot is at the station; when the robot is pulled away from the station by hand or through the wearable interface, it turns off within 2 s; when the robot is returned to the correct position, it turns on again within 5 s. No external measuring instrument is required for evaluation.

Charging-status indicator:
A battery-icon or multi-level indicator may be used. The minimum acceptance condition is that visible "charging" information is provided while the actual charging circuit is active and disappears automatically when the robot leaves the station.

4.4 Occupational Health, Safety, and Safe Working Practices

Batteries, power electronics, soldering equipment, hand tools, cutting/heating tools, and manufacturing equipment used within the project must be operated in accordance with the relevant laboratory and occupational health and safety rules. Each team is responsible for the safe use of the materials and equipment it uses, taking the necessary protective measures, and stopping work and notifying the advisor in the event of a hazardous condition. This technical specification does not replace the institution's applicable occupational health and safety procedures.

5. Module-Based Minimum Requirements

5.1 Wearable IMU/EMG Glove

- The glove must generate five different commands from the operator's hand/arm movement or muscle activity: forward, backward, turn left, turn right, and stop.
- Commands must be transmitted wirelessly to the robot; body-mounted buttons or manual commands from the phone may not be used during the acceptance test.
- User calibration may be performed for each command; however, the threshold/algorithm may not be changed after the final test.
- If electrical biosignals are to be measured on a person, only safe, battery-powered setups that are galvanically isolated from the mains may be used.

5.2 Mobile Camera-Based Evaluation Application

The mobile application must monitor the robot's task using the smartphone's own camera and its wireless communication with the robot, display basic status information to the user, and report the results automatically.

- Before the test, the phone is placed in a fixed position from which the robot's motion area is visible, while the robot is positioned balanced and stationary.
- The user confirms through the application that a wireless connection with the robot has been established, and the application begins presenting charging-status and tilt-angle information received from the robot. The application receives this information from the robot at 1 Hz. The connection must not be interrupted for more than 2 s while the application is running.
- For the start of the test, the user selects the task to be performed from the interface.

Tasks to Be Tested:

Task Name	Task Module	Success Metric
Static Balance	Module A	Remain balanced for 30 s
Robot Guidance	Module B	Correct direction and distance traveled (0.8-1.2 m range)
Charging	Module C	Detection of charging start within 5 s
Integrated Task	Module E	Successful transport of the medication box

- Charging status is measured using the camera (the robot's position relative to the charging station), the robot's tilt (its docking state at the charging station), and battery-level information.
- When the user presses the "Start Test" button, that moment is accepted as the start timestamp of the test duration and the application starts camera recording.
- During the task, the robot's position and motion are tracked by the camera while charging and tilt information received from the robot are displayed live on the interface. The interface also measures elapsed time with a timer.
- The robot's motion state (moving, turning, static balance, recovery/rebalancing) and direction information (forward, backward, right, left) are presented to the user live through the interface.
- When the task is completed or the time expires, the user presses the "End Test" button, or the application automatically detects the predefined completion condition and terminates the test.
- At this point, the application automatically determines whether the selected task has been completed successfully based on the measured duration (±2 s tolerance), distance traveled (0.8-1.2 m range) and motion direction, the robot's final position, and the condition of the transported item.
- The result is stored together with a video or timestamped record, the measured values, and the pass/fail decision in a form that cannot later be manually altered.

5.3 Standard Task and Load

The standard task load is one medication box (e.g., aspirin). The box must be closed, dry, and non-hazardous. The brand, medication content, or medical properties of the box are not scored; only whether it is transported safely is evaluated.

Recommended integrated task
The robot waits balanced for 5 s → moves approximately 1 m and changes direction using glove commands → completes the task without dropping the load → reaches the defined approach region of the charging station → from this point onward autonomously finds the station, aligns, and docks → automatic charging begins → the mobile application produces the test record and pass/fail result.

6. Recommended Materials List and Estimated Budget

The list does not prescribe any brand/model and its use is not mandatory. Teams may select equivalent or different parts, change quantities, and add extra features. It is assumed that the student already has a smartphone; the phone cost is not included.

Material group	Quantity	Estimated cost (TRY)
Robot microcontroller/development board	1	300-700
Robot IMU/tilt sensor	1	150-350
Geared drive motor	2	700-1.400
Bidirectional motor driver	1 set	250-650
Wheel, hub, coupling	2 sets	350-700
Battery pack	1	600-1.200

Material group	Quantity	Estimated cost (TRY)
BMS, regulator, fuse, switch, charging circuit	1 set	500-1.000
Automatic charging station + robot charging connection + status indicator + IR transmitter-receiver sensor	1 set	700-1.500
Wearable glove: MCU + IMU or EMG + battery + connections	1 set	800-2.200
Chassis/body and fabrication	1 set	600-1.200
Load carrier + medication box	1 set	100-300
Wireless communication module	2 sets	200-800
Cable, connector, solder/PCB, mounting elements	1 set	400-900
Other	-	500

Budget scale:
Estimated total: approximately TRY 6,150-13,400 per team. The typical target budget is approximately TRY 9,500. Laboratory hand tools, multimeter, soldering station, and use of available 3D printer/CNC facilities are not included in this amount. More expensive hardware does not increase the technical score.

HARDWARE SUPPORT: Student groups that request it may be provided, as a grant for use in project work, with a Raspberry Pi Model B or Raspberry Pi 2 Model B board together with a compatible power adapter (SD card not included). Use of these boards is not mandatory and does not directly contribute to the technical score.

7. Common Final Test Protocol

- The test is conducted on a flat, hard, dry, and level indoor surface.
- The robot carries the standard medication box in all tests; if the box falls, the corresponding subtest fails.
- Contact of any robot part other than the two main wheels with the floor, or loss of balance caused by a person touching the robot, causes the corresponding subtest to fail.
- For wearable commands, the physical response must begin within 2 s.
- For time measurement, the jury/advisor's stopwatch is the reference. No precise physical marker is required for approximately 1 m of motion; the mobile application performs the measurement using the phone's own camera and sensors.
- Each subtest is scored only as "pass = 5" or "fail = 0"; no partial score is awarded.

8. 100-Point Project Success Evaluation Table

Each of the five main criteria is worth 20 points and consists of four binary subtests worth 5 points each. The selected method, brand, appearance, or cost does not directly earn points.

Code	Criterion	Points	Definitive test	Scoring
C1	**Balance and basic motion**	**20**	1) 30 s static balance 2) Move approximately 1 m forward and stop for 3 s. 3) Return approximately 1 m backward and stop for 3 s. 4) Recover within 10 s from both approximately +10° and -10° initial tilts; both directions must be satisfied together.	Each item = 5 points; 0 if the condition is not fully satisfied.
C2	**Commands with wearable glove**	**20**	Five separate commands: forward, backward, turn left, turn right, and stop. For forward/backward commands, the robot moves approximately 0.5 m; for turning commands, it turns approximately 90°. The stop command is executed after all motion commands. The stop command is evaluated together with the other four commands and has no separate evaluation score. Each command comes wirelessly only from the wearable interface; the robot responds within 2 s and maintains balance.	Each successful command = 5; otherwise = 0.
C3	**Automatic charging station**	**20**	1) From the starting position within the defined 30 cm-radius and 90° approach region relative to the charging station, the robot detects and heads toward the station without human or wearable-interface guidance. 2) During the approach, the robot automatically adjusts its direction and speed and successfully docks in the charging position. 3) After correct docking, charging starts within 5 s without human intervention. The hardware charging indicator turns on automatically. 4) When the robot is pulled away, the indicator turns off within 2 s. When the robot is returned to the correct position, it turns on again within 5 s.	Each satisfied condition = 5; otherwise = 0.
C4	**Mobile visual measurement and evaluation**	**20**	1) The application establishes wireless communication with the robot and receives charging status and current tilt-angle information; these data are displayed live on the interface and communication occurs at 1 Hz. 2) The robot's motion and direction are detected from the camera and displayed to the user in real time. 3) According to the task selected from the interface, whether the robot successfully completes the task is evaluated automatically. 4) Each test automatically produces a results report with video or a timestamped record; the result cannot later be manually changed to "pass."	Each satisfied condition = 5; otherwise = 0.
C5	**Integrated task and repeatability**	**20**	The same integrated task is performed independently 4 times: the robot starts with the medication box; travels to the target point using wearable-interface commands; waits for 5 s without dropping the box; heads toward the charging station; starts charging without human intervention; and the mobile application produces a result record.	Each fully completed task = 5; otherwise = 0.

Technical success score:
C1 + C2 + C3 + C4 + C5 = 0-100. This table standardizes the team's technical project output; reflecting each student's individual contribution in the course grade is at the advisor's discretion.

8.1 Evaluation Record Form

Criterion	Subtest 1	Subtest 2	Subtest 3	Subtest 4	Total
C1 Balance and motion	0/5	0/5	0/5	0/5	__/20
C2 Glove	0/5	0/5	0/5	0/5	__/20
C3 Automatic charging	0/5	0/5	0/5	0/5	__/20
C4 Mobile application	0/5	0/5	0/5	0/5	__/20
C5 Integrated task	0/5	0/5	0/5	0/5	__/20
OVERALL					__/100

9. Project Summary and Minimum Requirements

Project topic: Two-Wheeled Indoor Patient Assistance Robot

Objective: To develop a two-wheeled robot that actively maintains its own balance, transports a standard medication box to an indoor target location using commands from a wearable wireless interface, reaches the automatic charging station on its own wheels at the end of the task and begins charging without human intervention, and has its task measured and reported by an application using the smartphone camera/sensors.

Mandatory modules: Balancing robot, wearable IMU/EMG or equivalent human-machine interface, automatic charging station and hardware charging-status indicator, mobile camera/sensor-based evaluation application, and medication-box transport.

Areas of freedom: Control method, mechanical dimensions and exterior design, sensor/microcontroller/motor selection, IMU/EMG or another suitable human input, wireless communication, phone-based image/sensor processing method, programming language, charging-station contact/energy-transfer method, manufacturing method, and features added beyond the minimum requirements.

Fundamental fairness and safety rules: The robot may not remain upright passively; no third support may be used in scored tests; a ready-to-operate balancing robot may not be used as the main platform; charging may not be initiated by a person connecting a cable or manually placing the robot on the station; the charging indicator must be connected to the actual charging circuit; all work must be carried out in accordance with applicable laboratory and occupational health and safety rules.

10. Recommended Workflow for the Semester

Time	Expected work
Weeks 1-2	Team, scenario, role allocation, and system architecture.
Weeks 2-4	Part selection, mechanical/power design, glove, and mobile-application framework.
Weeks 4-7	Robot subsystems and initial active balancing; automatic charging-station and glove prototypes.
Weeks 7-10	Motion control, wearable commands, mobile visual tracking, and measurement.
Weeks 10-12	Medication-box transport, charging-station docking, integrated task.
Weeks 12-13	Rehearsal using the common test protocol, troubleshooting, and result recording.
Final/Project Fair	Live 5×20 evaluation, integrated demonstration, and team presentation.

Final principle:
The specification fixes the minimum requirements, not the solution. Teams are encouraged to develop different mechanical designs, different control methods, different wearable human-machine interfaces, different automatic charging solutions, and additional features; evaluation is based only on common, measurable outcomes.

Budget note: Planning ranges were updated to be consistent with the Turkish hobby/robotics retail price scale as of 23.09.2026. No brand/model is mandatory.